

EEG Based Biometric Authentication System

Prince Yadav

Department of Electrical Engineering
Indian Institute of technology, Palakkad
Email: 122401029@smail.iitpkd.ac.in

Abstract—Electroencephalogram (EEG) signals provide unique neural patterns that can be used for secure biometric authentication systems. This project presents an EEG-based multi-user biometric authentication system using Digital Signal Processing (DSP) techniques and a Convolutional Neural Network (CNN). The EEG signals are preprocessed using normalization and a Butterworth Bandpass Filter to remove unwanted noise and preserve relevant brainwave frequencies. Continuous Wavelet Transform (CWT) is then applied to generate Scalogram images representing the time-frequency characteristics of the EEG signal. These scalograms are divided into smaller patches and used as input to a CNN for feature learning and classification. The system supports multi-user authentication and includes unknown-user rejection using majority voting. The proposed method is evaluated using standard performance metrics such as Accuracy, FAR, FRR, Precision, Recall, F1-Score, MCC, and Confusion Matrix. Experimental results demonstrate the effectiveness of the proposed EEG authentication system.

Index Terms—EEG, Biometric Authentication, Digital Signal Processing, CNN, Scalogram, Wavelet Transform, Bandpass Filter, Deep Learning

I. MOTIVATION

Traditional authentication techniques such as passwords, PINs, and fingerprint-based systems are widely used in security applications. However, these methods can still face issues such as password leakage, spoofing attacks, and duplication of biometric information. EEG-based authentication provides an additional level of security because brain activity patterns are difficult to imitate or reproduce artificially. Recent improvements in signal processing and deep learning techniques have also made EEG analysis more practical for biometric applications. These factors motivated the development of an EEG-based authentication framework.

II. OBJECTIVE

The primary objective of this project is to develop a EEG based biometric authentication system using digital signal processing and deep learning techniques. The work focuses on preprocessing EEG signals, extracting useful time-frequency information using Wavelet Transform, generating scalogram images, and classifying users using a CNN model. Another important objective was to include unauthorized-user rejection so that the system could differentiate between registered and unknown users and improve performance by decreasing FAR and FRR.

III. LITERATURE REVIEW

A. Development of an EEG-Based Biometric Authentication Security System

In 2015, Hasan Mubin proposed an EEG-based biometric authentication framework that demonstrated the feasibility of using brain signals for secure user authentication. The system included signal preprocessing, feature extraction, and classification using a Support Vector Machine (SVM). The study showed that EEG signals possess unique characteristics that can be used to reliably identify users and improve security compared to traditional biometric systems. The work established a complete EEG authentication pipeline and highlighted the potential of EEG as a robust biometric modality.

B. Survey of EEG-Based Biometric Authentication

In 2016, Isuru Jayarathne, Michael Cohen, and Senaka Amarakeerthi presented a comprehensive survey on EEG-based biometric authentication systems. The study reviewed different EEG preprocessing techniques, feature extraction methods, and machine learning classifiers such as SVM and Artificial Neural Networks (ANN). The survey concluded that EEG signals are highly secure because brainwave patterns are difficult to replicate or forge. It also discussed challenges such as signal variability, noise handling, and channel selection in EEG-based authentication systems.

C. EEG-Based Biometric Identification with Deep Learning

In 2017, Zijng Mao, Wan Yao, and Yufei Huang proposed an EEG-based biometric identification system using deep learning techniques. The system converted EEG signals into image-like representations and used Convolutional Neural Networks (CNNs) for automatic feature learning and classification. The proposed deep learning approach achieved high identification accuracy and demonstrated that CNNs can effectively learn discriminative EEG features without manual feature engineering.

D. EEG-Based Person Identification using CNN

In 2023, Waleed Alsumari et al. proposed an EEG-based person identification and authentication system using Deep Convolutional Neural Networks. The system achieved very high identification accuracy using only two EEG channels and short-duration EEG recordings. The study demonstrated that CNN-based architectures significantly improve EEG authentication performance and are suitable for practical real-world biometric applications.

IV. DATABASE DESCRIPTION

The dataset used in this project consists of EEG recordings stored in EDF (European Data Format) files. Each EEG recording contains multiple channels collected using scalp electrodes placed at different locations on the head.

The EEG recordings were segmented into fixed-length epochs before further processing. Separate EEG samples were used for training and testing purposes. Additional unseen subjects were also included during evaluation to analyze the unknown-user rejection capability of the system.

Dataset Details:

- Dataset Name: EEG Motor Movement/Imagery Dataset
- Number of Users: 109
- Sampling Frequency: 160 Hz
- Number of EEG Channels: 64 EEG Electrodes
- Data Nature: Time-series voltage signals (in microvolts)
- Each subject performed 14 experimental runs
- Fixed Segment or Epoch Duration: 2 seconds

V. SYSTEM ARCHITECTURE

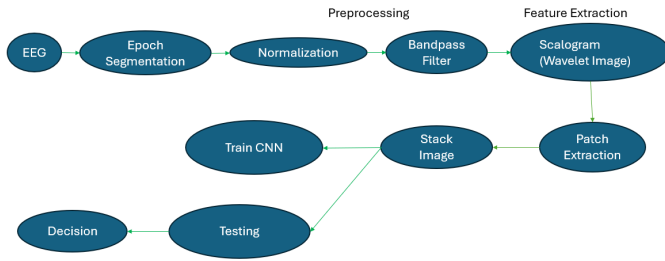


Fig. 1. System Block Diagram

The complete processing pipeline consists of the following stages:

- 1) EEG Acquisition
- 2) Epoch Segmentation
- 3) Signal Normalization
- 4) Signal Filtering
- 5) Wavelet Transform
- 6) Scalogram Generation
- 7) Patch Extraction
- 8) CNN Classification
- 9) Authentication Decision

A. EEG Preprocessing

1) **Normalization:** The raw EEG signals were first divided into fixed-length epochs of 2 seconds. Channel-wise Z-score normalization was then applied using the following equation:

$$X_{norm} = \frac{X - \mu}{\sigma}$$

where:

- X = EEG signal

- μ = Mean value
- σ = Standard deviation

Normalization helped reduce amplitude variations between EEG channels and improved training stability during CNN classification.

2) **Signal Filtering:** Different filtering techniques, such as low-pass, high-pass, notch, and bandpass filters, were analyzed during experimentation.

Butterworth IIR filters were selected because they provide a smooth frequency response and preserve important EEG characteristics effectively.

Normalized cutoff frequencies are calculated using the Nyquist frequency:

$$W_n = \frac{f_c}{f_N}$$

where:

- f_c = cutoff frequency
- $f_N = \frac{f_s}{2}$ = Nyquist frequency

The general transfer function of an IIR filter is given by:

$$H(z) = \frac{b_0 + b_1z^{-1} + b_2z^{-2} + \dots + b_Mz^{-M}}{1 + a_1z^{-1} + a_2z^{-2} + \dots + a_Nz^{-N}}$$

Low-pass Filter:

A 4th-order Butterworth Low-Pass Filter with a cutoff frequency of 30 Hz was applied to the EEG signal. The low-pass filter removes high-frequency noise while preserving low-frequency EEG components.

$$H(z) = \frac{0.038 + 0.1519z^{-1} + 0.2278z^{-2} + 0.1519z^{-3} + 0.038z^{-4}}{1 - 0.9784z^{-1} + 0.7901z^{-2} - 0.2419z^{-3} + 0.0377z^{-4}}$$

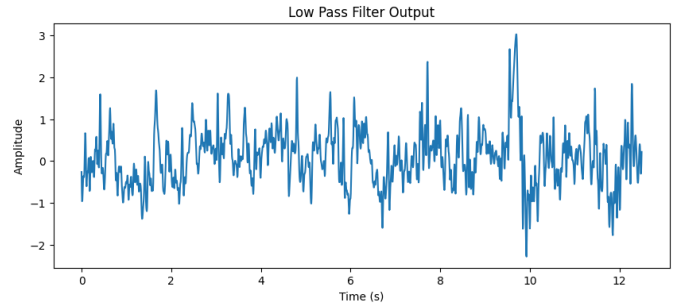


Fig. 2. Low Pass Filtered Signal

High-pass Filter:

A 4th-order Butterworth High-Pass Filter with a cutoff frequency of 0.5 Hz was implemented. The high-pass filter suppresses low-frequency drift and baseline fluctuations in EEG recordings.

$$H(z) = \frac{0.9747 - 3.8987z^{-1} + 5.848z^{-2} - 3.8987z^{-3} + 0.9747z^{-4}}{1 - 3.9487z^{-1} + 5.8474z^{-2} - 3.8487z^{-3} + 0.95z^{-4}}$$

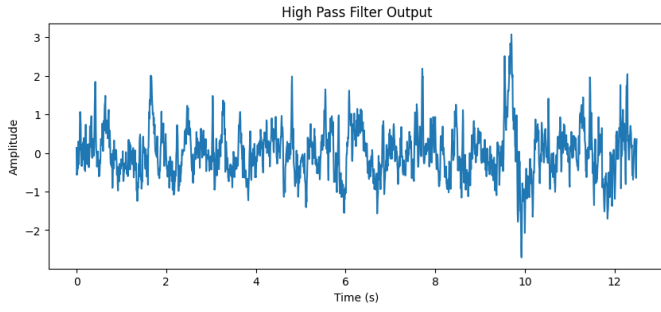


Fig. 3. High Pass Filtered Signal

Bandpass Filter:

A 4th-order Butterworth Band-pass Filter with cutoff frequencies of 8 Hz and 30 Hz was finally selected. As Most useful EEG rhythms (α and β bands) are concentrated within this frequency range.

During initial experimentation, wider frequency ranges introduced additional noise and reduced classification consistency. The selected 8 to 30 Hz range provided comparatively stable results.

$$H(z) = \frac{0.114 - 0.228z^{-2} + 0.114z^{-4}}{1 - 2.3002z^{-1} + 2.2474z^{-2} - 1.1692z^{-3} + 0.3007z^{-4}}$$

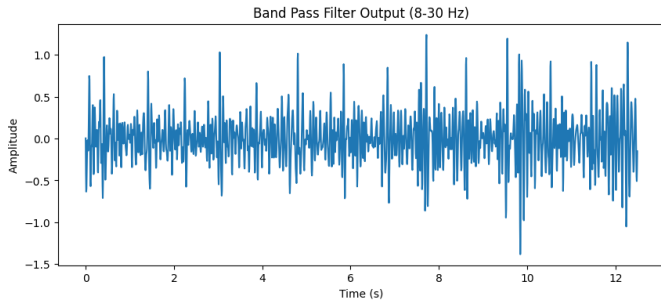


Fig. 4. Band Pass filtered signal

Notch Filter:

The notch filter was analyzed for removing power-line interference around 50 Hz.

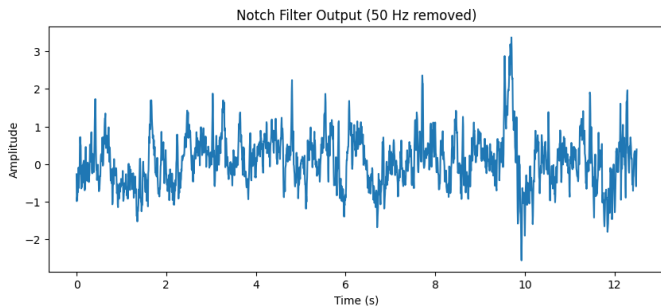


Fig. 5. Notch Filter

Filter Comparison:

Among all filtering methods analyzed, the band-pass filter provided the best balance between noise reduction and EEG signal preservation. So band-pass filter is selected for this project.

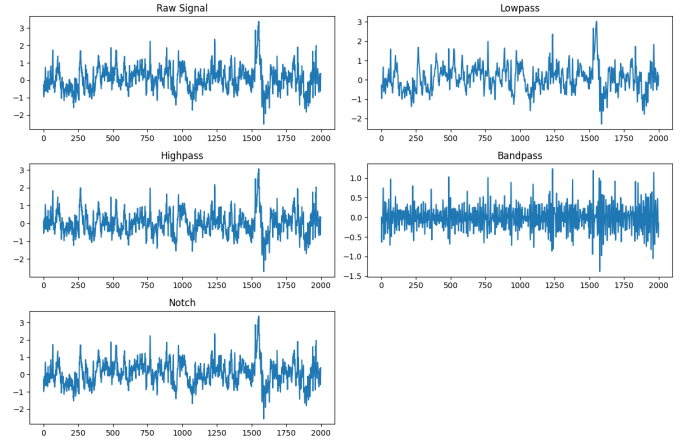


Fig. 6. Comparing Different Filters

B. Feature Extraction

Different feature extraction techniques such as FFT, Spectrogram, and Wavelet Transform were analyzed before selecting the final approach.

1) **Fast Fourier Transform (FFT)**: FFT converts EEG signals from the time domain into the frequency domain.

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N}$$

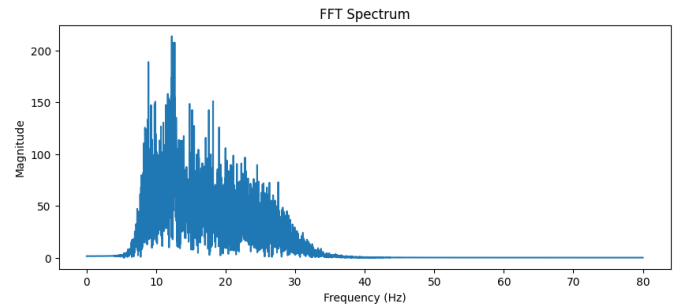


Fig. 7. FFT Spectrum

FFT was initially tested because of its simplicity, but it could not preserve temporal EEG variations effectively and Shows frequency content only no time localization.

2) **Spectrogram (STFT)**: The Short-Time Fourier Transform (STFT) provides combined time-frequency analysis but have fixed resolution so not suitable for non-stationary EEG signals.

$$STFT\{x(n)\}(m, \omega) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n}$$

where:

- $x[n]$ = Input Signal
- $w[n]$ = window function
- m = time shift

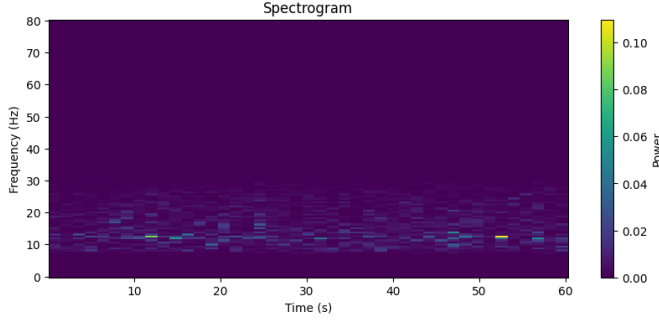


Fig. 8. Spectrogram

Although STFT provided additional temporal information, the spectrograms were comparatively more sensitive to noise.

3) **Wavelet Transform:** Continuous Wavelet Transform (CWT) was finally selected because it performs multi-resolution time-frequency analysis efficiently for non-stationary EEG signals and captures transient changes.

$$CWT(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt$$

where:

- a = scale parameter
- b = translation parameter
- ψ = mother wavelet

4) **Scalogram:** Wavelet coefficients were converted into scalogram images representing EEG energy distribution across both time and frequency.

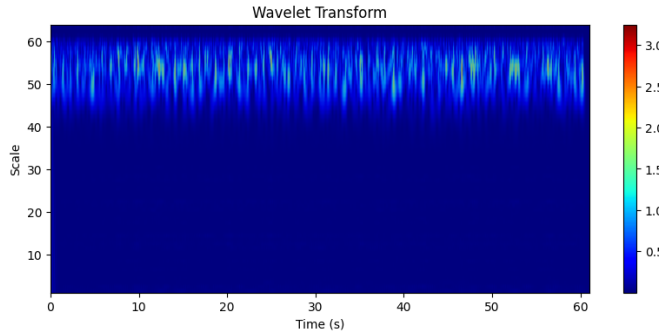


Fig. 9. Scalogram

Compared to FFT and STFT outputs, the scalogram representation preserved richer EEG characteristics and improved CNN feature learning during experimentation.

C. Patch Extraction

Instead of directly resizing the generated scalograms, the images were divided into smaller fixed-size patches. This approach helped preserve local EEG information and reduced feature loss.

The extracted patches from multiple EEG channels were stacked together and used as CNN input tensors.

VI. CNN MODEL AND CLASSIFICATION

The generated scalogram patches were supplied to a Convolutional Neural Network (CNN) for feature learning and user classification.

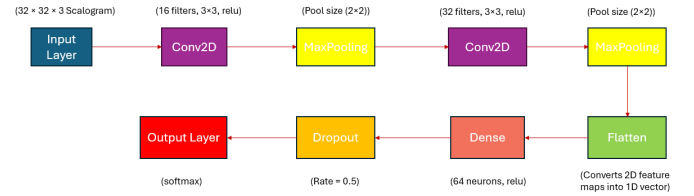


Fig. 10. CNN Architecture

A. CNN Architecture

The implemented CNN architecture consists of the following layers:

- Convolution Layer
- ReLU Activation
- Max Pooling Layer
- Flatten Layer
- Fully Connected Layer
- Dropout Layer
- Softmax Output Layer

The CNN automatically learned discriminative EEG patterns from the generated scalogram patches.

B. Final Hyperparameters

TABLE I
FINAL HYPERPARAMETERS

Hyperparameter	Value
Input Size	$32 \times 32 \times 3$
Optimizer	Adam
Loss Function	Sparse Categorical Crossentropy
Batch Size	16
Epochs	30
Dropout Rate	0.5
Activation Function	ReLU
Output Activation	Softmax
Kernel Size	3×3
Pooling Size	2×2

VII. RESULTS AND PERFORMANCE ANALYSIS

The developed EEG authentication model was evaluated using standard classification metrics.

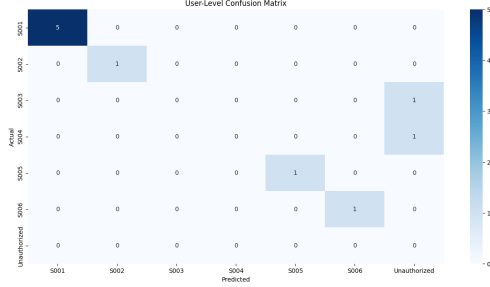


Fig. 11. User-Level Confusion Matrix

A. Confusion Matrix

- True Positive (TP): Authorized user correctly authenticated
- True Negative (TN): Unauthorized user correctly rejected
- False Positive (FP): Unauthorized user accepted
- False Negative (FN): Authorized user rejected

B. Accuracy:

Measures overall correctness of the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

C. Precision:

Measures how many predicted snores are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

D. Recall (Sensitivity):

How many authorized users are correctly accepted

$$Recall = \frac{TP}{TP + FN}$$

E. Specificity:

How well we reject imposters.

$$Specificity = \frac{TN}{TN + FP}$$

F. F1-Score:

The harmonic mean of Precision and Recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

G. False Acceptance Rate (FAR):

Unauthorised user wrongly accepted

$$FAR = \frac{FP}{FP + TN}$$

H. False Rejection Rate (FRR):

Authorised user wrongly rejected

$$FRR = \frac{FN}{FN + TP}$$

I. Matthews Correlation Coefficient (MCC):

Measures the overall quality of binary classification by considering all four confusion matrix values

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

J. Latency:

Latency represents the average time required for authenticating a single EEG sample.

TABLE II
PERFORMANCE METRICS

Metric	Value
Accuracy	0.87
Precision	0.79
Recall	0.76
Specificity	0.96
F1-Score	0.77
FAR	0.04
FRR	0.23
MCC	0.76
AUC-ROC	0.97
Latency	0 sec

VIII. OBSERVATIONS

The implemented EEG authentication framework combined signal processing and CNN-based deep learning techniques for multi-user biometric authentication.

Among the tested filtering methods, the bandpass filter preserved useful EEG rhythms more effectively while suppressing unwanted noise components.

Wavelet-based scalogram representations produced better feature quality compared to FFT and STFT representations because both temporal and spectral information were preserved simultaneously.

The CNN model successfully learned discriminative EEG patterns from the generated scalograms. It was also observed that majority voting improved authentication consistency during unknown-user testing.

IX. FINAL CONCLUSION

In this project, an EEG-based biometric authentication system was implemented using digital signal processing and deep learning methods.

The developed pipeline included EEG preprocessing, filtering, wavelet-based feature extraction, scalogram generation, and CNN-based classification. The model was capable of

authenticating multiple users while also detecting unauthorized users.

Experimental analysis indicates that EEG signals can be effectively used for secure biometric authentication applications. Future improvements can include larger datasets, reduced computation time, and real-time system implementation.

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